roles. However, they can also be specified in terms of their behavior modeled as a set of rules, e.g., whether an agent usually responds to emails very briefly or circumlocutory, or by relationships to others, e.g., who is the boss of whom. These properties should be utilized for corresponding actions executed by the knowledge worker and for the task assignment.

Moreover, the users can specify general simulation settings in advance, e.g., how many tasks the agents should complete or the probability that an agent will get sick. In the current prototype, the focus was on defining tasks to shape the simulation. In the next subsection, we explain how tasks are designed and how they can be specified by the KnoWoGen users.

Tasks: Tasks are a central element of the simulation and specify the activities of the knowledge workers. In our design, tasks are sequences of actions with logical or content dependencies, i.e., a meeting invitation has to be written and sent out before the actual meeting happens. Examples of tasks are, for instance, preparing for a workshop or writing a paper. Hence, tasks comprise multiple activities, here called actions, like writing a section for the paper or giving feedback to a colleague's paper. We base our definitions of action types on the taxonomy of Reinhardt et al. [Re11] defining types such as authoring, dissemination, and information search.

Users of the KnoWoGen have to specify the tasks in the configuration [4]. Tasks are defined by their frequency, which actions they comprise, and other general properties relevant to all actions, like the topic of the task, which agents should be involved, and other involved entities. For instance, a project or product subject to a task can be defined. Actions are characterized by their duration, their action type, and other type-specific properties, e.g., for an authoring task, the document type that should be written must be set. Moreover, pointers to actions, a certain action depends on, have to be given such that these dependencies can be considered by the simulation engine.

The tasks guide the simulation process. In contrast to other typical Multi-agent Systems (MAS), the actions are predefined and not chosen based on environmental observation. Nevertheless, the output of actions still shapes the environment, and previously generated documents can be considered when producing others. For specific actions, generated documents can be analyzed and follow-up actions can be derived. For instance, this is the case for email conversations since an email can contain questions that should be answered by the receiver and should spark an action of writing a response. Another difference to other MAS is that there is also no assessment phase of the actions' outputs because the objective of the KnoWoGen is not necessarily optimally solving knowledge work tasks. Instead, the goal is to generate authentic documents and data traces which are difficult to check automatically.

[4] In the current prototype, the configuration can be specified in a TOML file

Most actions result in a document since the desired knowledge work dataset should be document-centric. There are, however, also actions that only result in data traces, e.g., when putting a document in a folder or searching.

Generating Documents: Documents are created when knowledge workers execute actions assigned to them. They represent the outcome of completed actions. Depending on the action configuration, a prompt is composed and sent to an instruction-fine-tuned LLM that generates a suitable document based on the given instructions. Hereby, relevant parameters of the simulation environment are utilized as input, e.g., which agents are involved in the action, which topic should be discussed, and which document type should be generated. Those inputs either stem from the task configuration or are randomly sampled during the simulation setup. The prompts are composed of up to four parts that are listed and described in the following [5]:

- A system prompt part describes the general goal of generating an authentic artificial HTML-formatted document that might be filled with additional information not mentioned in the subsequent prompt parts.
- Depending on the action, an instruction part specific to the underlying action type and configuration, states, for instance, which agents are involved, which topic the document should address, and which type of document should be generated.
- Summaries of previous documents that should be considered are included.
- Further document type-specific instructions targeting problems found in pretests.

The choice of the concrete LLM connected to the KnoWoGen is part of the configuration. LLMs with higher context length limits are preferable since the instructions can be more detailed, and longer documents can be synthesized with one prompt. Depending on the chosen LLM, prompt templates tailored to a different LLM might have to be adapted minorly to achieve comparable results.

Filling the Knowledge Graph: Besides the environment setup, the contextual information of the simulation steps is stored in the knowledge graph. This includes, in particular, all details about tasks, e.g., all executed actions and their properties like their dependencies, involved agents, and other parameters. Moreover, all resulting data traces and documents are included and connected to their underlying actions.

The simulation context stored in the knowledge graph can later be used as input or ground truth depending on the tools that should be optimized or evaluated. For example, task information can be used as ground truth for a task predictor. Similarly, document content dependencies can be used for corresponding predictors or action parameters for classifiers.

[5] Concrete examples of prompts and corresponding, generated documents can be found here: https://purl.archive.org/knowogen/examples

4 Experiments

Similar to previous publications [Cl21; Ja23; JHN23], we examined how authentic generated documents, in this case knowledge work documents, are perceived by human evaluators compared to real ones. With this experiment, we assessed our approach's document generation capability. Document generation was, in the past, one main challenge of dataset generators and is an important factor for the general feasibility of our approach. Moreover, participants were asked in the experiment to explain their ratings to get more insights into the driving factors of their judgment and essential aspects of document authenticity.

Setup: The experiment dataset was composed of 25 emails and meeting minutes. We chose these two document types since they reflect or represent collaborative knowledge work. Hence, compared to other documents such as papers or project reports, they are associated with higher authenticity requirements since the collaboration or communication captured in the documents must also appear human-like.

To increase comparability, we selected five content categories for the included documents. Emails were either meeting invitations, retrospective meeting-related exchanges, or feedback on an external document. Included meeting minutes were discussions about the current work status. One category of meeting minutes involved also planning, i.e., a distribution of future tasks, and the other one focused solely on exchanging the status.

For each category, we collected one real example and generated four examples with the Llama-13B-Chat model [To23]. The real emails were randomly selected from the Enron dataset [KY04] and the meeting minutes from the ELITR dataset [Ne22]. Since we wanted to test two generation variables and for each one two different variants, we decided on a multivariate experiment. Generated documents were either more greedy or more deterministic and generated in a zero-shot or a two-shot manner, i.e., providing no or two examples of real documents in the prompt. For the two-shot variants, we also randomly sampled further examples from the two aforementioned datasets. All synthesized documents resulted from small simulations, i.e., they were the result of either the first or the second simulation step and accordingly only relied on at most one previously generated document.

We asked participants to rate on a 7-point Likert scale [Li32] how realistic the documents appeared to them. Additionally, they were encouraged to justify their decision. However, this was optional to avoid forcing participants to explain their decision when they were unsure about concrete influence factors. For each participant, documents were randomly ordered to decrease the effects of their order. Moreover, they were not told about the distribution of real and generated documents to not bias their judgments.

In total, 29 participants, aged between 18 and 64, rated the given knowledge work documents. Among them, most had a computer science background. Predominately, they were researchers (38%), students (31%), and software engineers (17%). 87% assessed their English proficiency